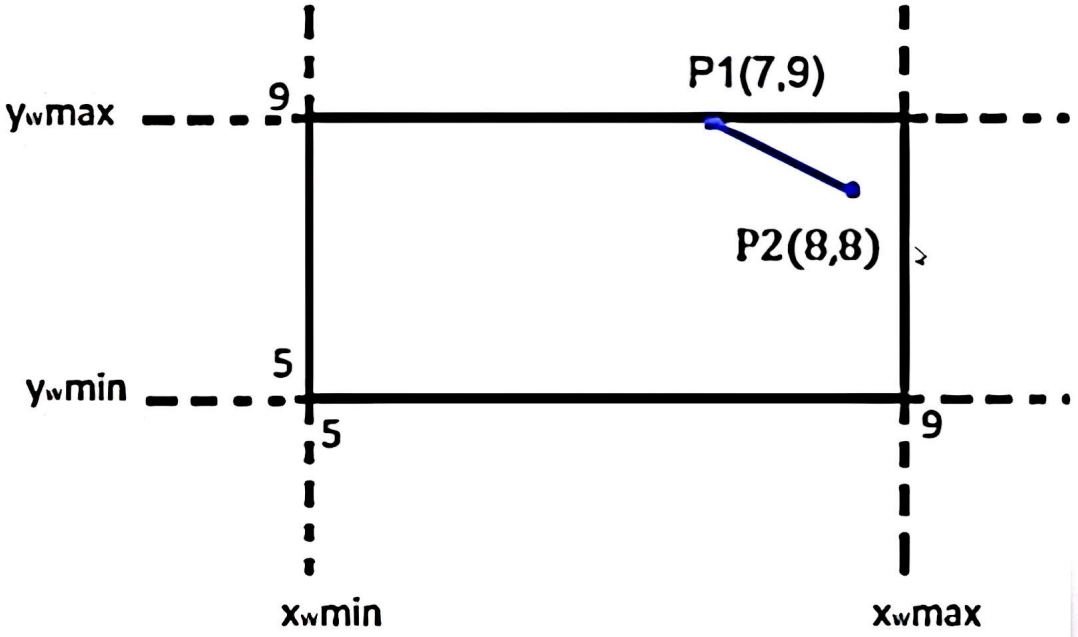
$$x = x_1 + \frac{(y - y_1)}{m}$$

$$= 5 + \frac{(9-11)}{-1} = 5 + 2 = 7$$

So the new point is (7,9)

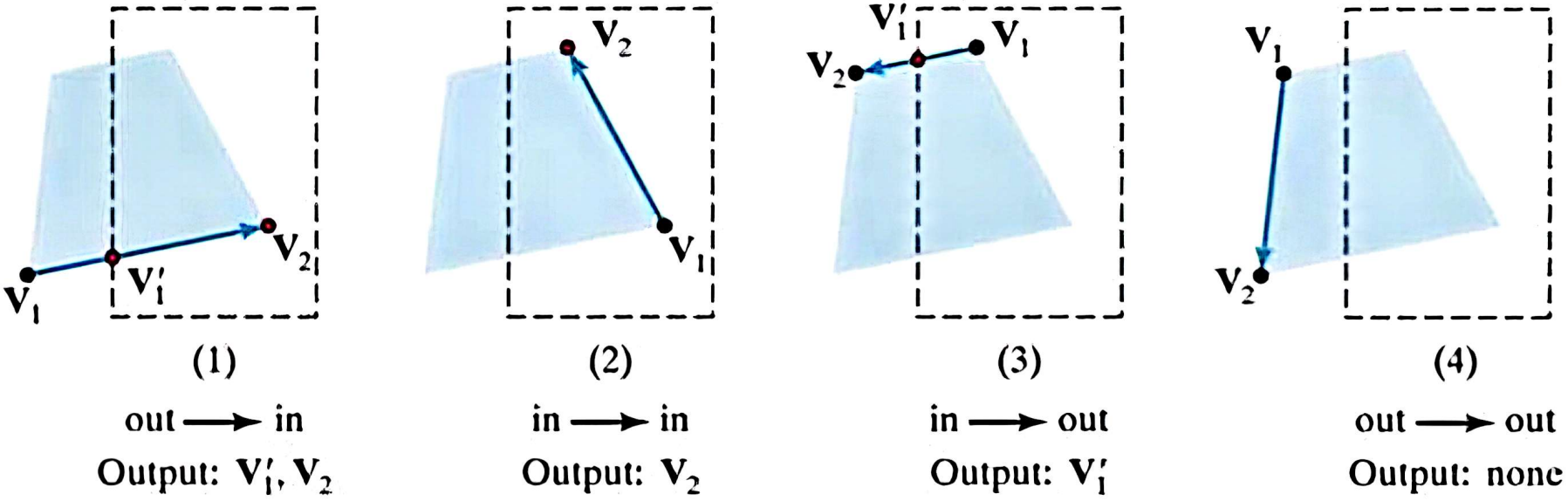
$x_w min \leq 7 \leq x_w max$
$y_w min \leq 9 \leq y_w max$

New Line Location: (7,9) to (8,8)

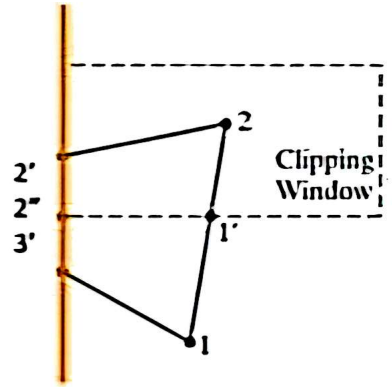


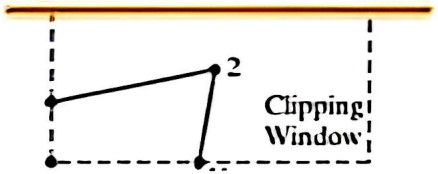
CamScanner

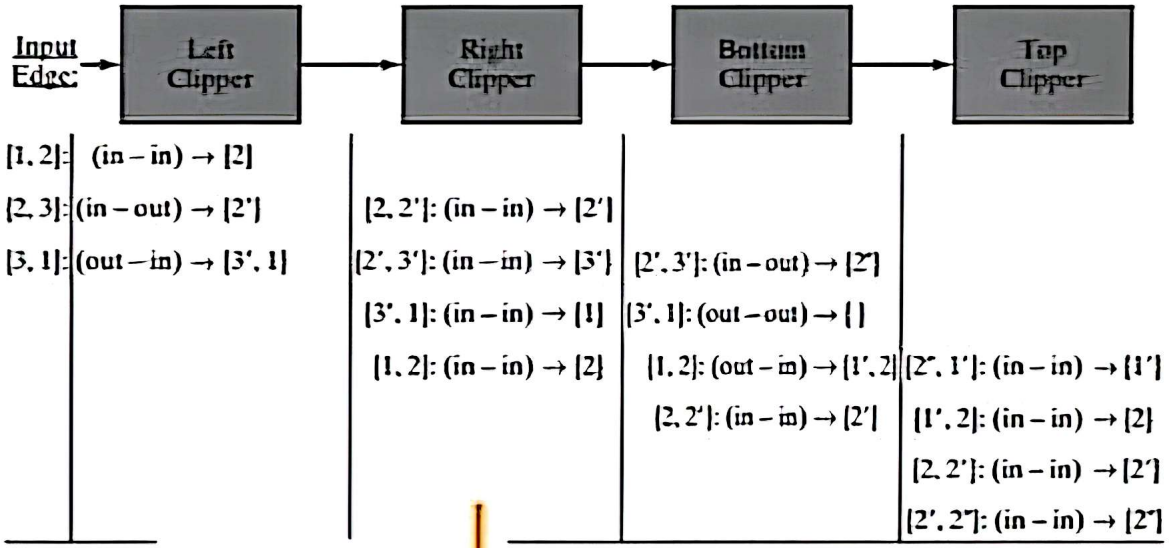
3-Sutherland--Hodgman Polygon Clipping

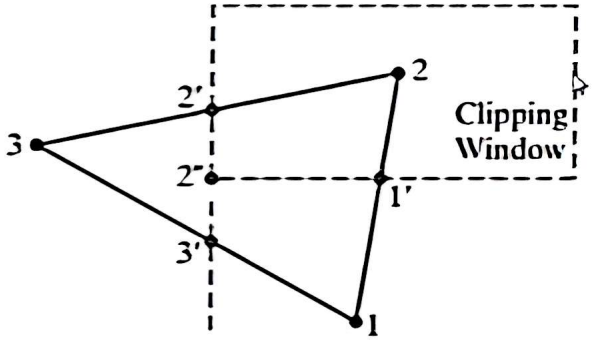


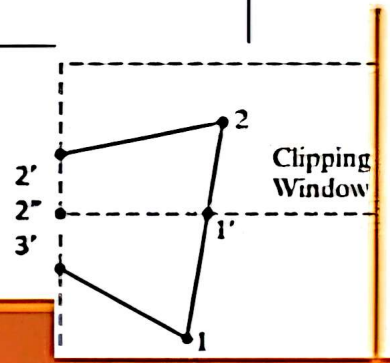
CamScanner

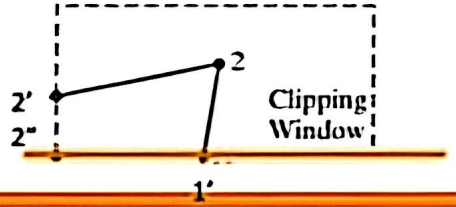

2
2'
2"
3'
1'
1
Clipping
Window


2
Clipping
Window


Input
Edge:
Left
Clipper
Right
Clipper
Bottom
Clipper
Top
Clipper
[1, 2]: (in – in) → [2]
[2, 3]: (in – out) → [2']
[3, 1]: (out – in) → [3', 1]
[2, 2']: (in – in) → [2']
[2', 3']: (in – in) → [3']
[3', 1]: (in – in) → [1]
[1, 2]: (in – in) → [2]
[2', 3']: (in – out) → [2"]
[3', 1]: (out – out) → { }
[1, 2]: (out – in) → [1', 2]
[2, 2']: (in – in) → [2']
[2", 1']: (in – in) → [1']
[1', 2]: (in – in) → [2]
[2, 2']: (in – in) → [2']
[2', 2"]: (in – in) → [2"]


3
2
2'
2"
3'
1'
1
Clipping
Window


2
2'
2"
3'
1'
1
Clipping
Window


2
2'
2"
Clipping
Window
1'

4-Weiler-Atherton Polygon Clipping

1. Process the edges of the polygon fill area in a counterclockwise order until an inside-outside pair of vertices is encountered for one of the clipping boundaries; that is, the first vertex of the polygon edge is inside the clip region and the second vertex is outside the clip region.

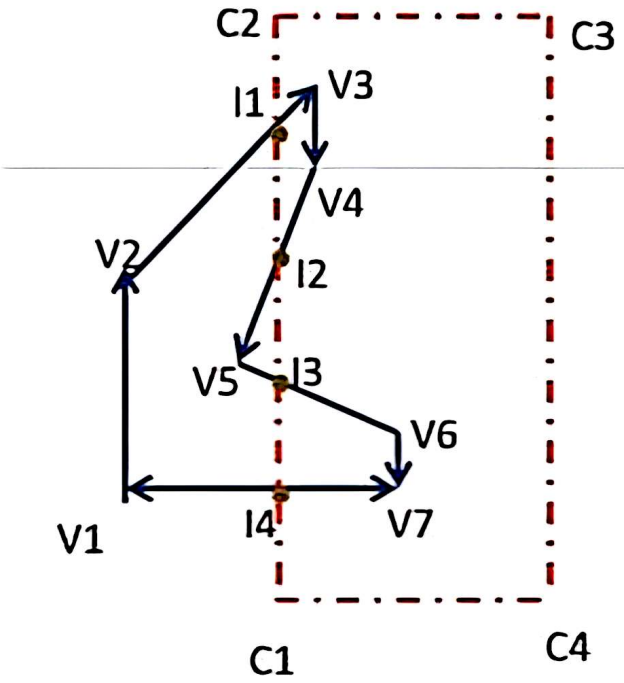
2. Follow the window boundaries in a counterclockwise direction from the exit-intersection point to another intersection point with the polygon. If this is a previously processed point, proceed to the next step. If this is a new intersection point, continue processing polygon edges in a counterclockwise order until a previously processed vertex is encountered.

3. Form the vertex list for this section of the clipped fill area.

4. Return to the exit-intersection point and continue processing the polygon edges in a counterclockwise order.

CamScanner

v....vertices i...intersection c ...clipped polygon

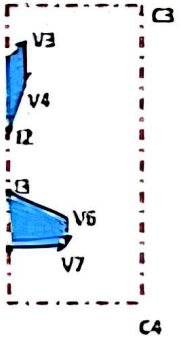


Subjected	clipping
V1	C1
V2	I4
I1 (OUT-IN)	I3
V3	I2
V4	I1
I2(IN-OUT)	C2
V5	C3
I3 (out-in)	C4
V6	
V7	
I4(in-out)	
V1	

Start from point (out-In)

End at point (In-out)(go to the next column

Start from point (out-In)



The First figure I1-V3-V4-I2-
The second figure I3-V6-V7-I4

CamScanner

Sec 8 : Filling

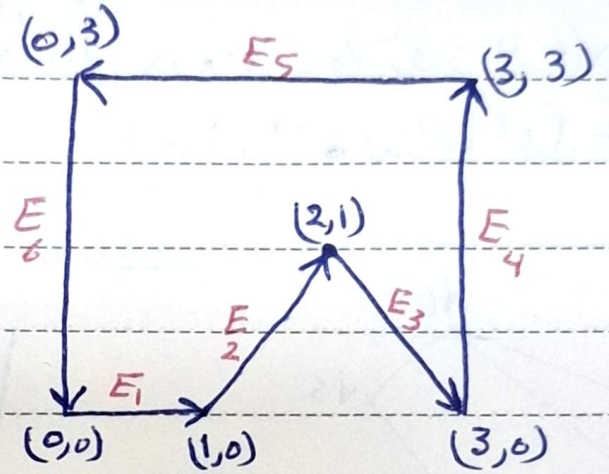
Note: Some graphics packages require all fill polygons to be Convex ⇒ Some accept only triangular fill areas

So we will split a ConCave into set of Convex

* to find the vector between 2 points

$E_1 \rightarrow (X_2 - X_1, y_2 - y_1, z_2 - z_1)$

EX1) find the vectors & checks if it's ConCave polygon



$E_1 = (1-0, 0-0, 0) = (1,0,0)$

$= (1, 0, 0)$

$E_1 \times E_2 = (0, 0, 1*1 - 0*1)$

$= (0, 0, 1)$

$E_2 = (1, 1, 0)$

$E_2 \times E_3 = (0, 0, 1*-1 - 1*1)$

$= (0, 0, -2)$ ∴ ConCave

$E_3 = (1, -1, 0)$

$E_3 \times E_4 = (0, 0, 1*3 - 0*-1)$

$= (0, 0, 3)$

$E_4 = (0, 3, 0)$

$E_4 \times E_5 = (0, 0, 0 - (-9))$

$(0, 0, 9)$

$E_5 = (-3, 0, 0)$

$E_5 \times E_6 = (0, 0, 9)$

$E_6 = (0, -3, 0)$

$E_6 \times E_1 = (0, 0, 0*0 - (-3*1))$

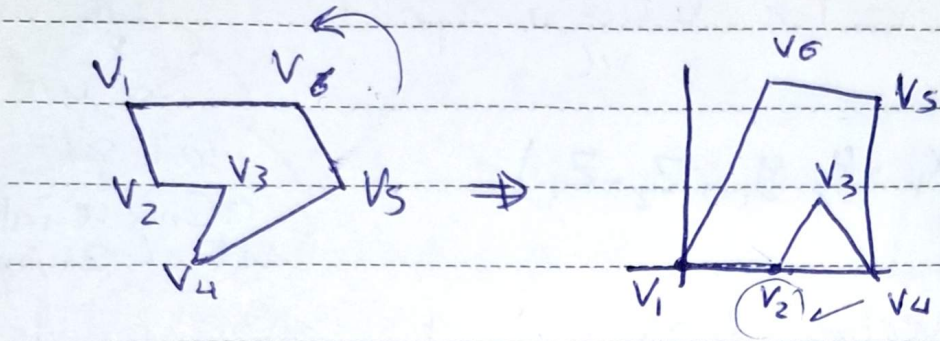
$= (0, 0, 3)$

Note: * the Cross product bet. 2 vectors (X,y) represent (Z) vector

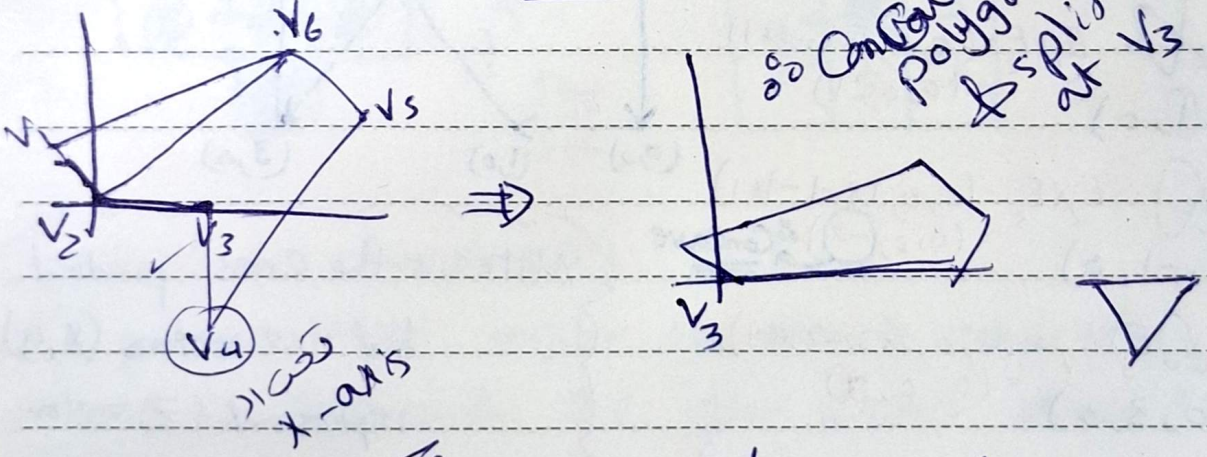
* to check if ConCave $E_i \times E_j < 0$ for any vector in the polygon

* Rotational method

نقل كل vertex الـ origin بحيث يكون الـ edge اللي بعده على الـ X-axis



نفس الشيء الـ polygon عند كل vertex ← واشوف الـ vertex اللي بعده فوق ولا تحت الـ X-axis لو تحته بيعده (Concave Polygon) وعندها اعمل split

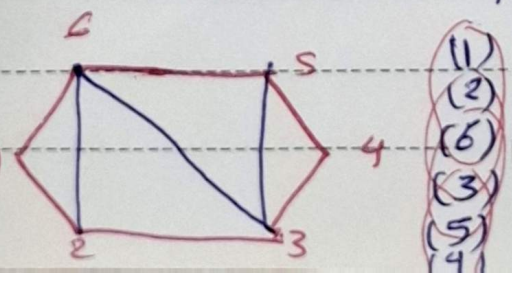


وبعدين اعمل Check لحد ما تأكد ان كل الزوايا اقل من 180

* Splitting Convex Polygon into set of triangles

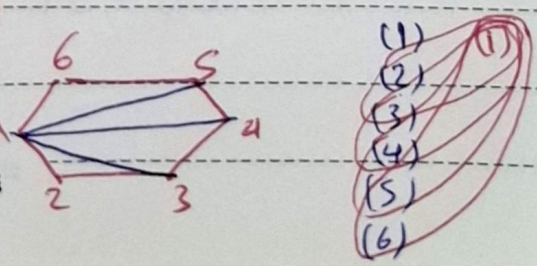
(GL-TRIANGLE-STRIP)

(Common edge) for N vertex we get (N-2) triangle



(GL-TRIANGLE-FAN)

(Common vertex) for N we get (N-2)



DAWENHA designs

CamScanner

Filling algorithms :

1 Boundary fill : (edge fill)

① Start at a point inside the polygon boundary

② Check on the neighboring pixel if → not colored & have not boundry color ↳ then color it

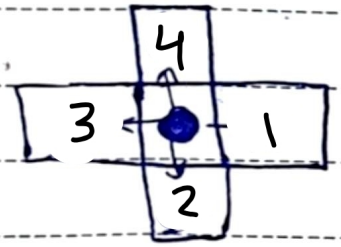
* only work if filled & boundry color are different

③ Store its neighboring pixels 4N or 8N in the stack

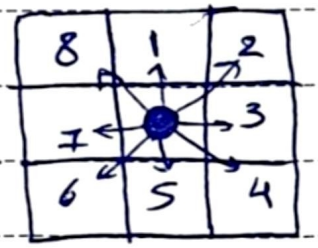
④ Select next pixel from stack & repeat ②



4Neighbors

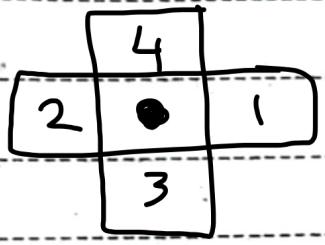


8N

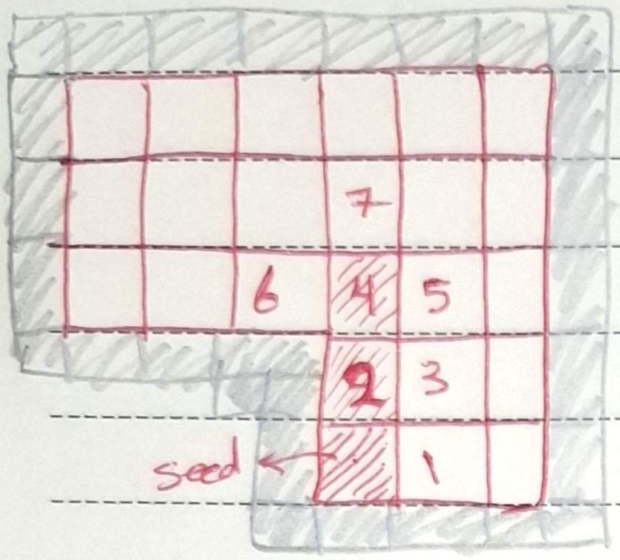


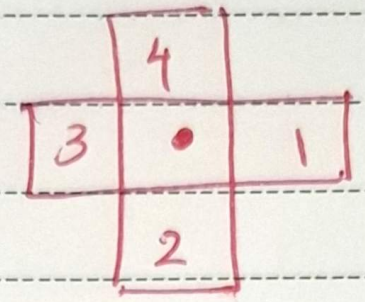
2 Flood fill (seed fill)

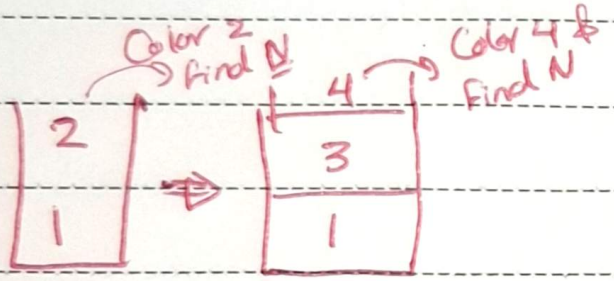
if you want to recolor the area it replace the interior color with fill color & start from an interior point (seed point)

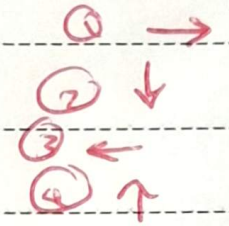


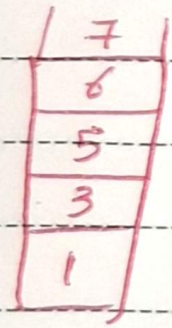
using boundry filling Show the stack to fill the polygon using 4 N



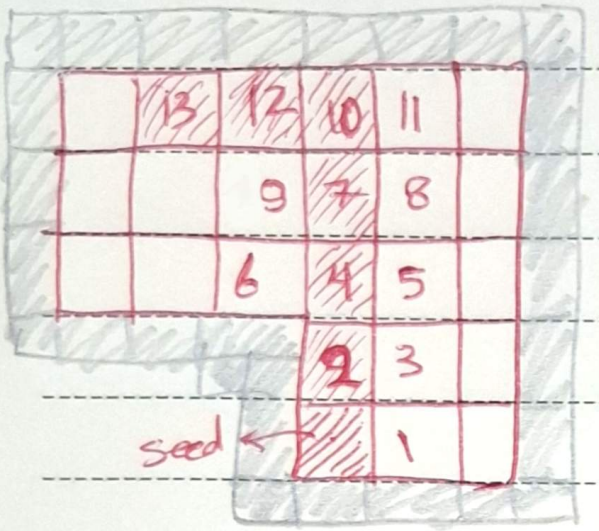


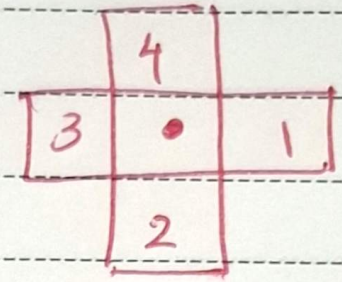


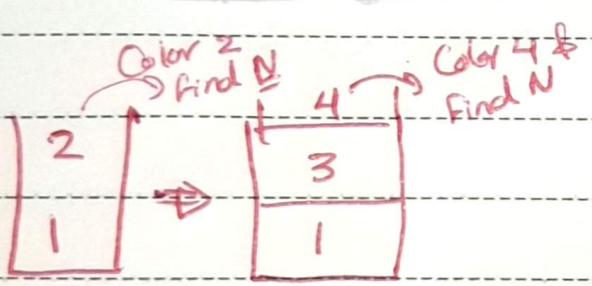


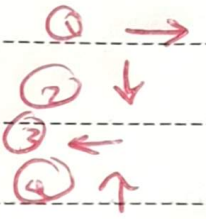


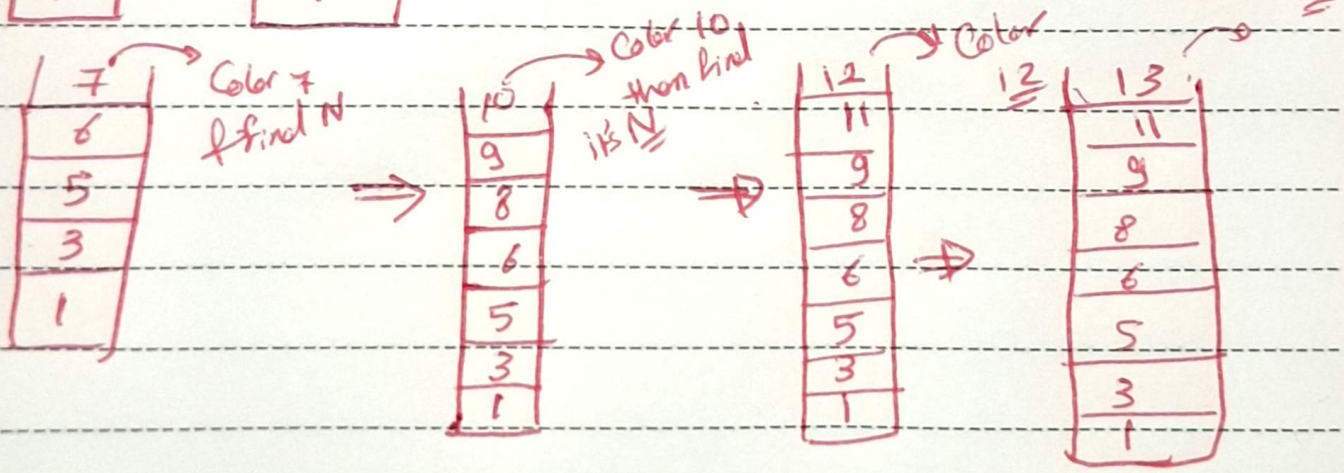
Using boundry filling Show the stack to fill the polygon using 4 N





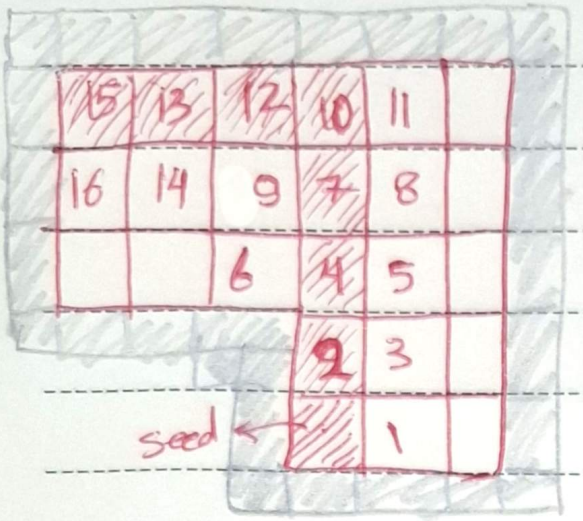


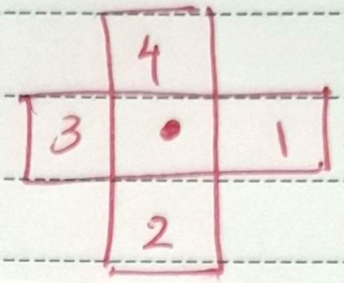


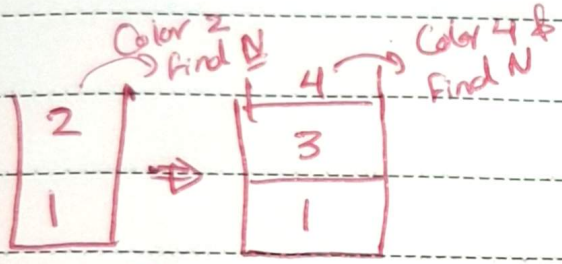


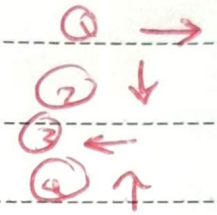
DAWENHA

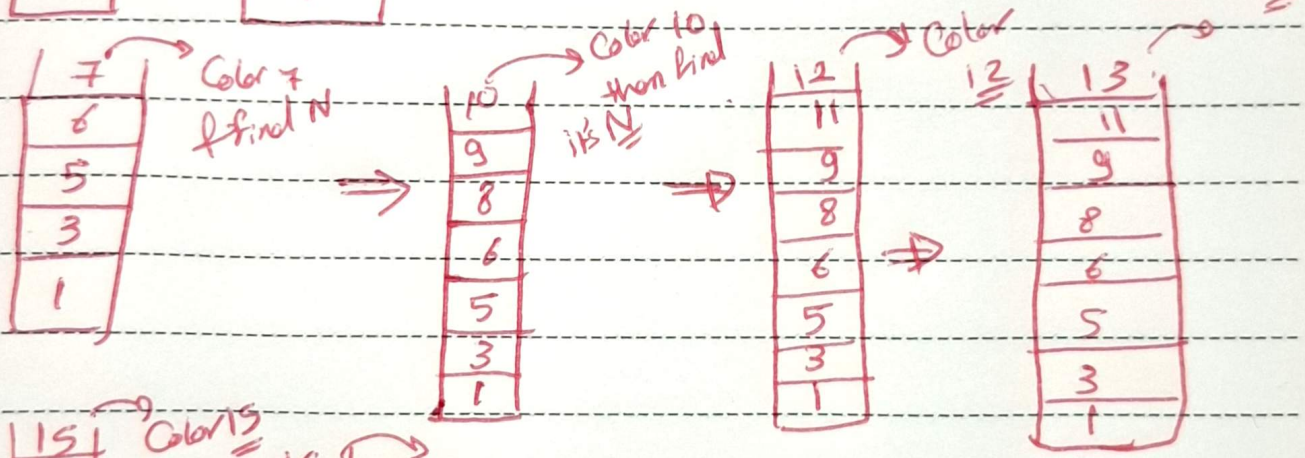
Using boundry filling Show the stack to fill the polygon Using 4 N

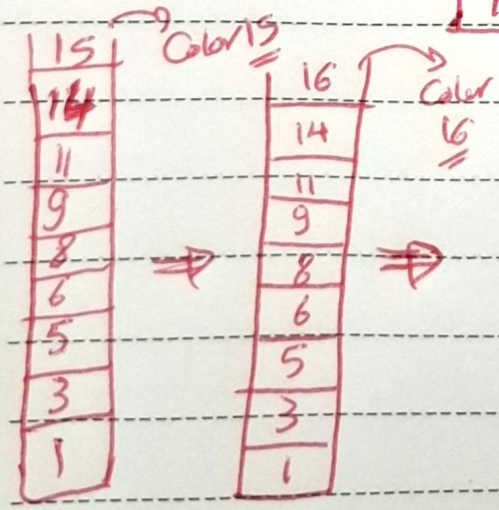






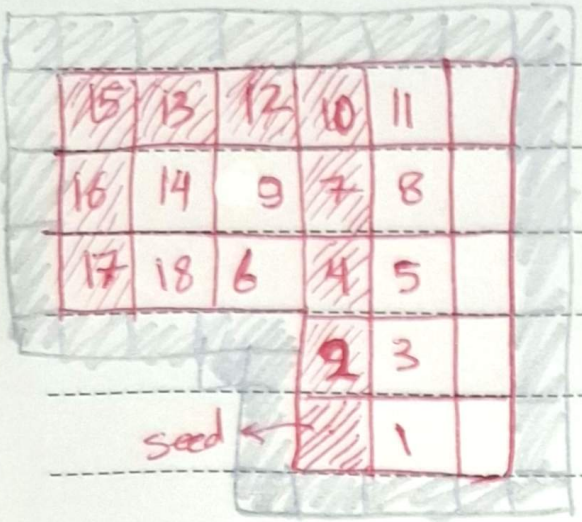


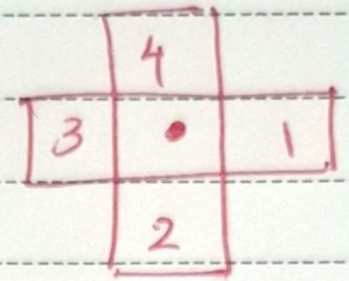


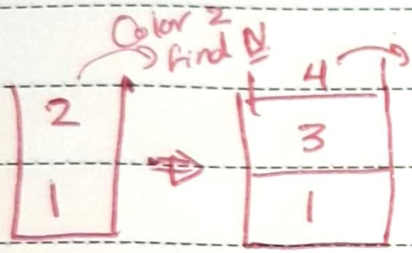


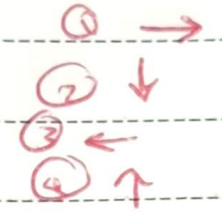
DAWENHA designs

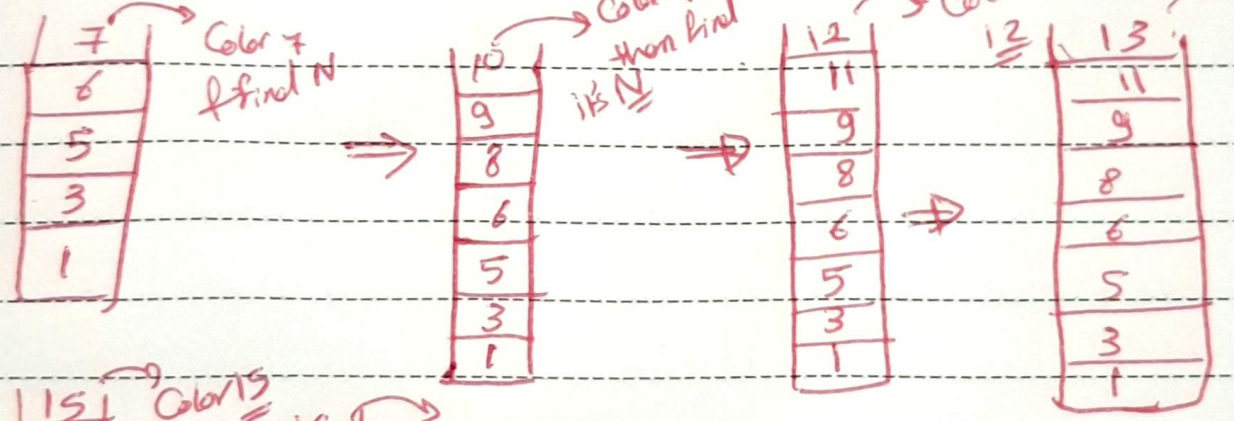
Using boundry filling Show the stack to fill the polygon using 4 N

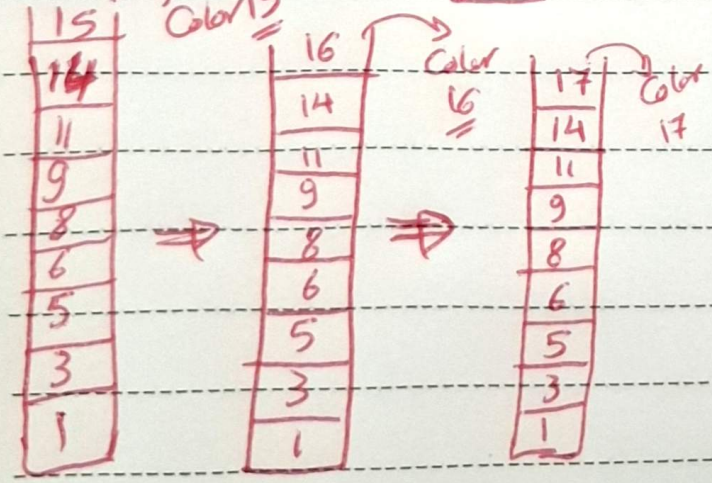








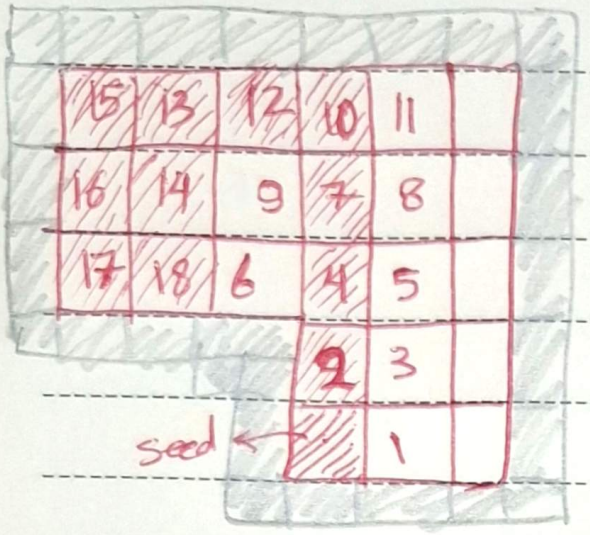


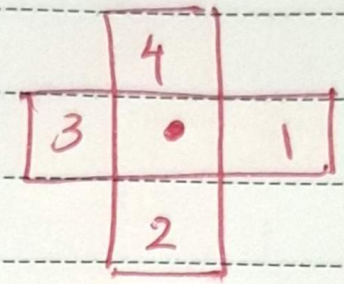


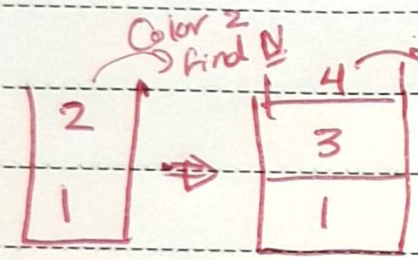
DAWENHA designs

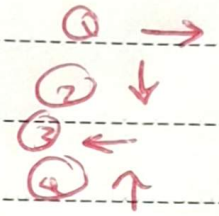
CamScanner

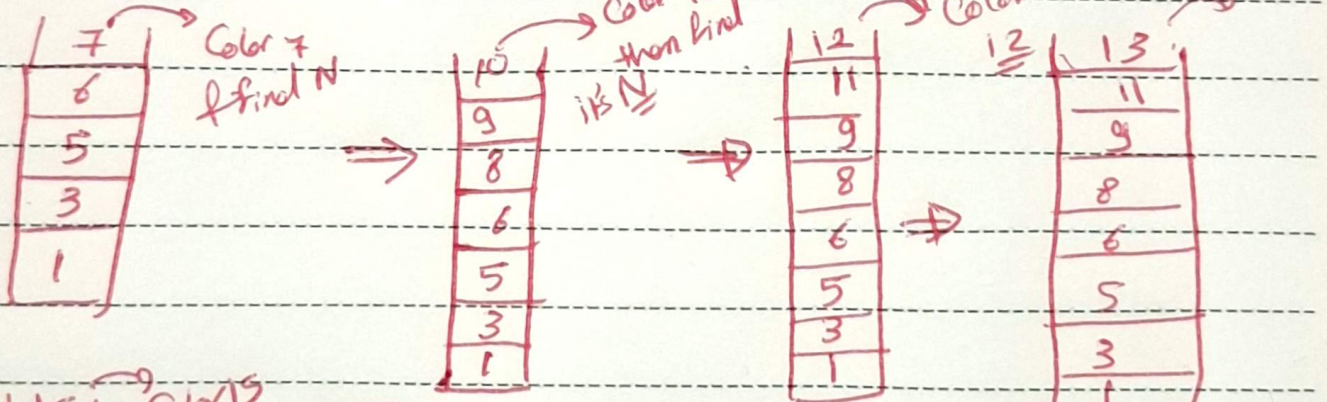
Using boundry filling Show the stack to fill the polygon using 4 N

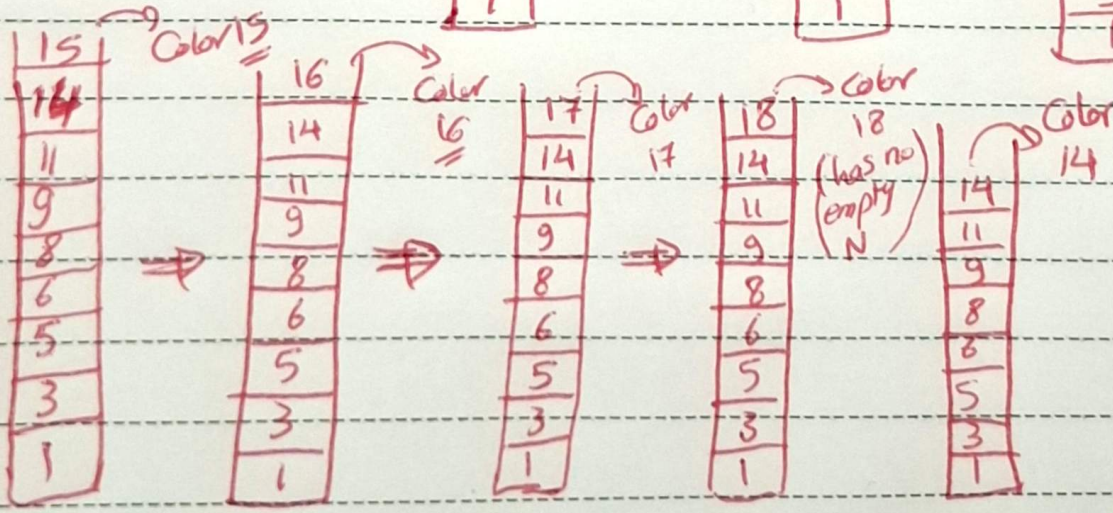








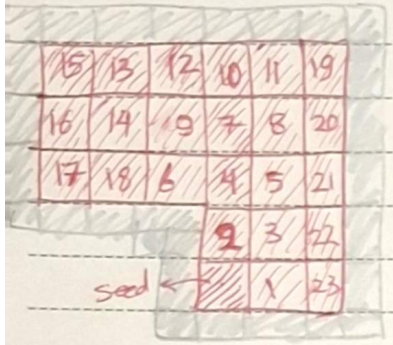


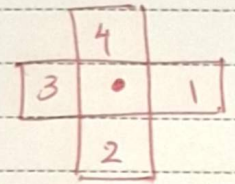


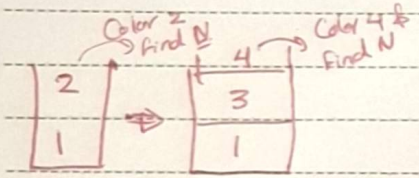
DAWENHA designs

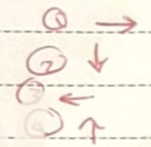
CamScanner

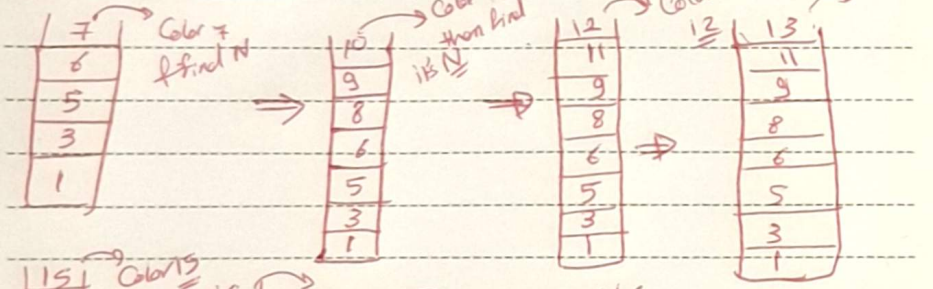
Using boundry filling Show the stack to fill the polygon using 4 N

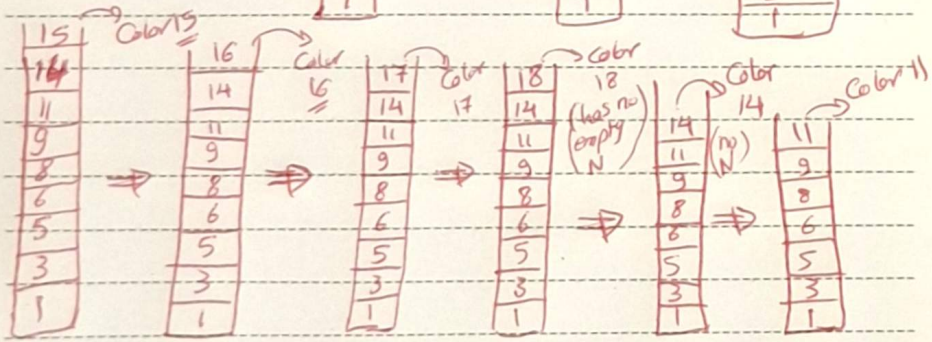




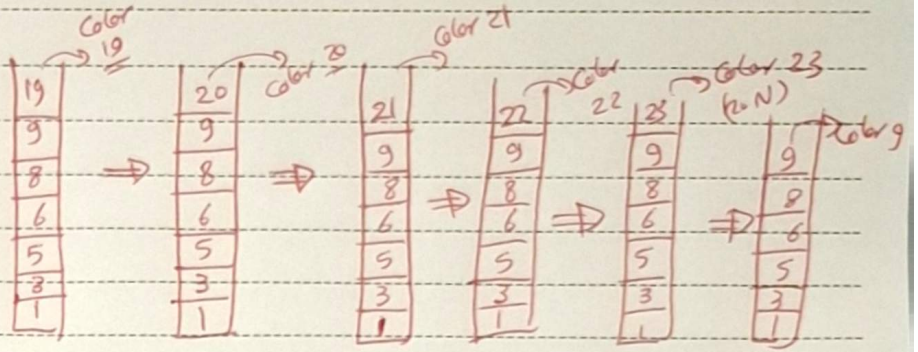


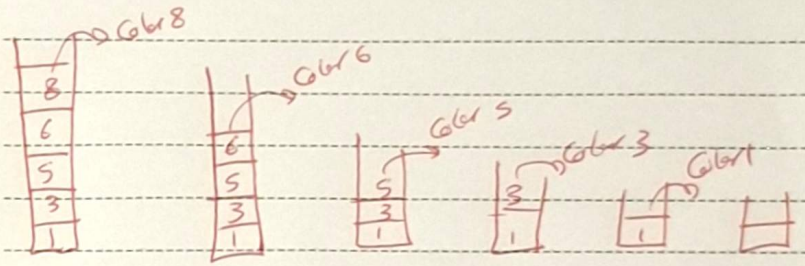






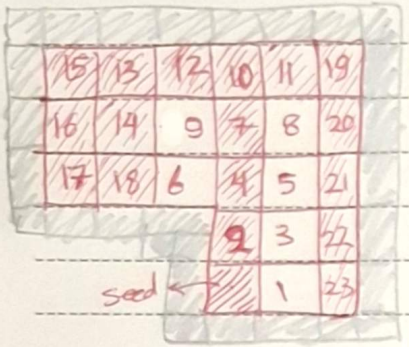
DAWENHA designs

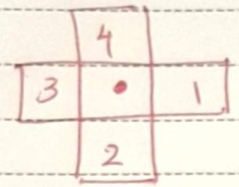


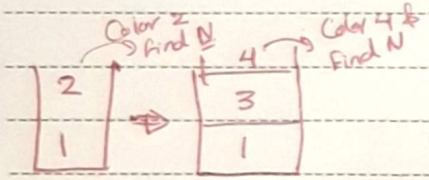


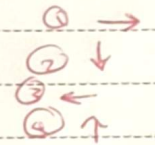
DAWENHA designs

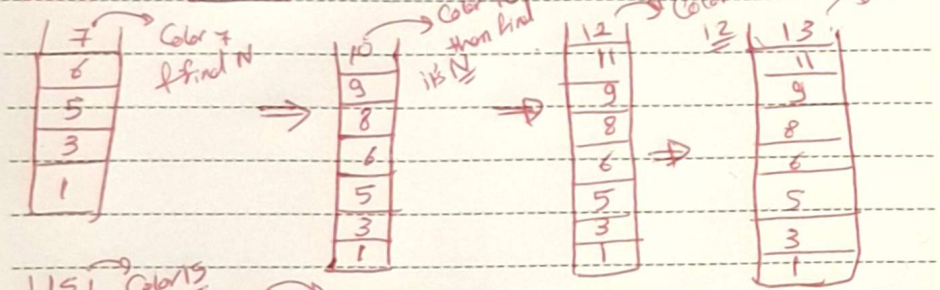
Using boundry filling Show the stack to fill the polygon using 4 N

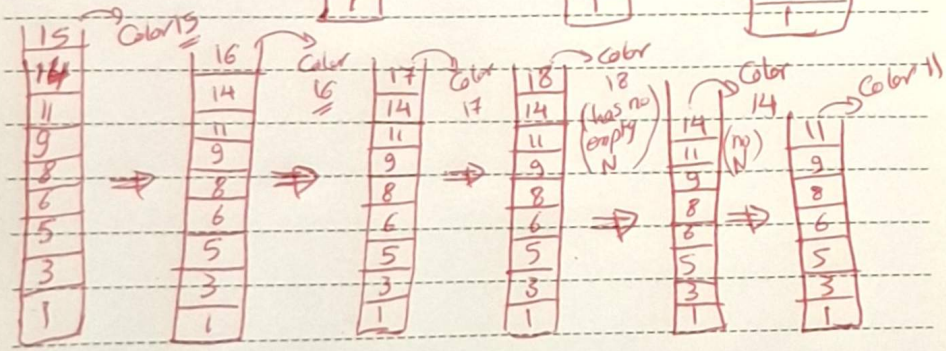




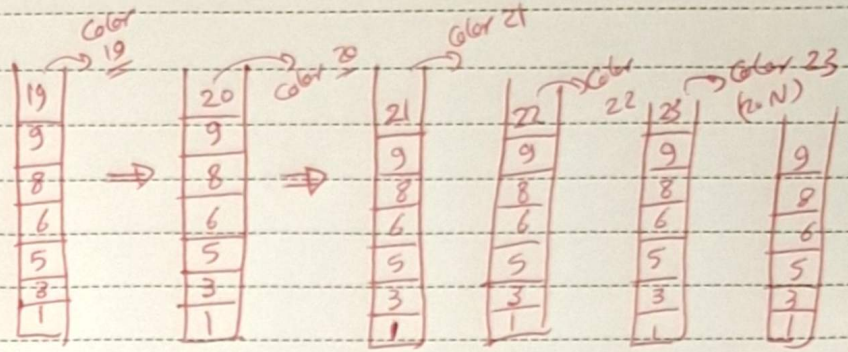








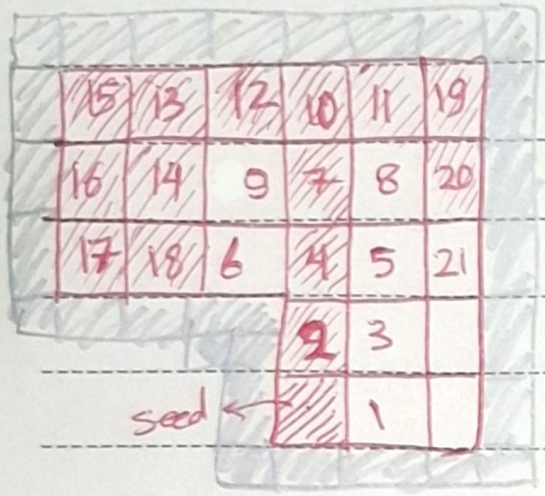
DAWENHA designs

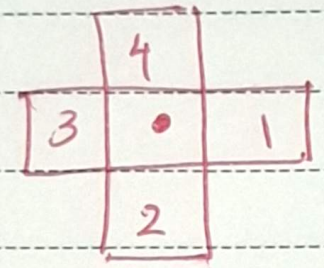


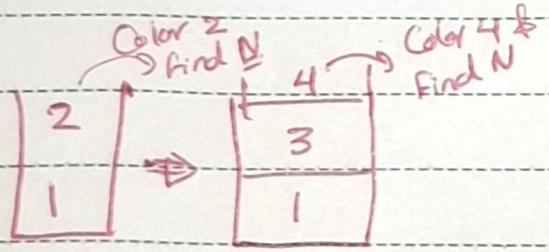
DAWENHA designs

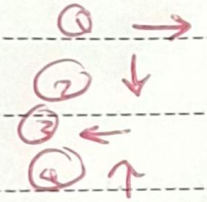
CamScanner

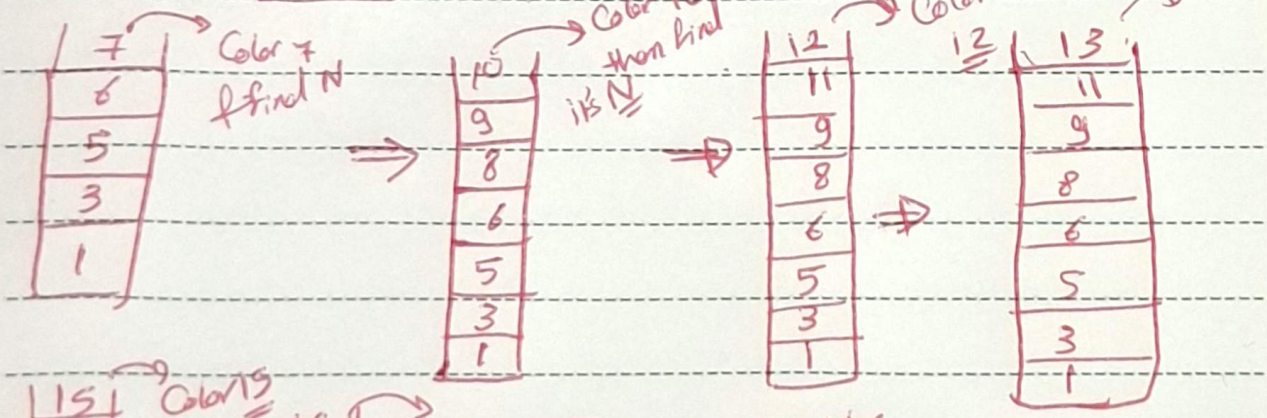
Using boundry filling Show the stack to fill the polygon using 4 N

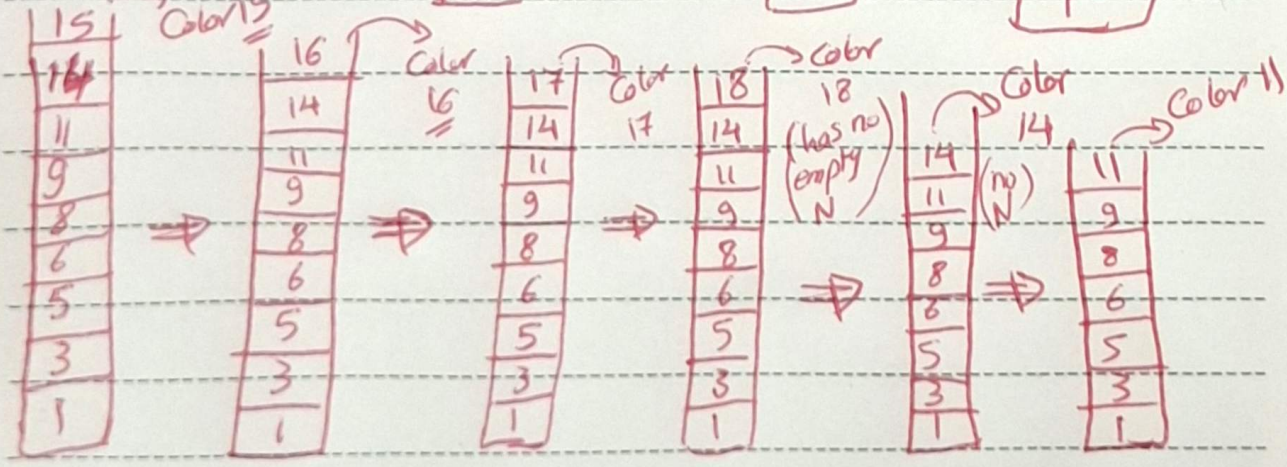








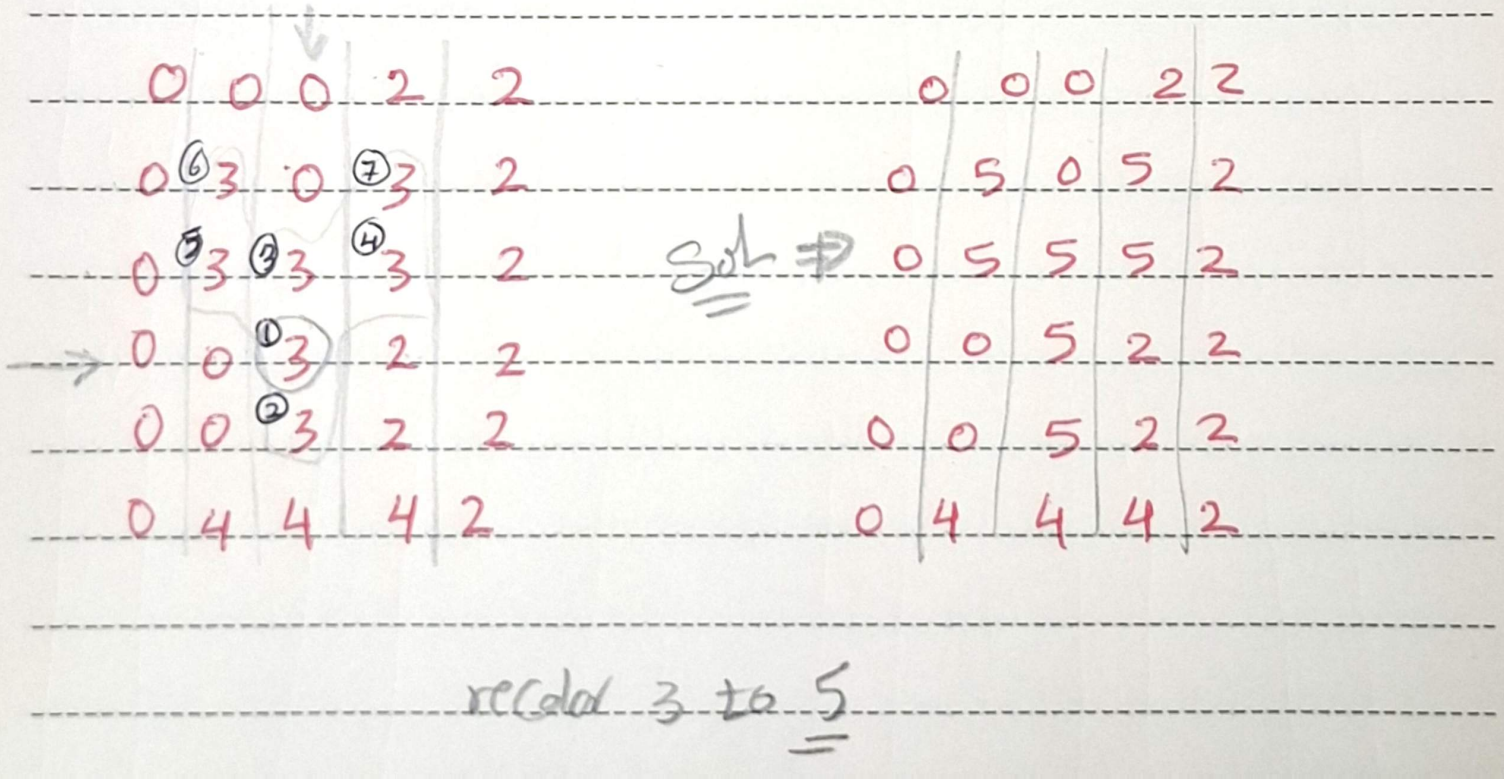




DAWENHA designs

Ex. using flood filling if the starting pixel Contain the interior Color is (3,2) (the 4th row & 3rd column) & th fill Color is 5

Show the grids after filling using 4N way Connectivity.



recolor 3 to 5

* winding-number rule (non-zero) : Can be used to define interior regions (Can unambiguously handel complex polygon including self-intersections & nested polygons)

① set w-n = 0 --- initial value of winding no

② draw line from point p to outside (vertex لازم يكون بره الشكل)

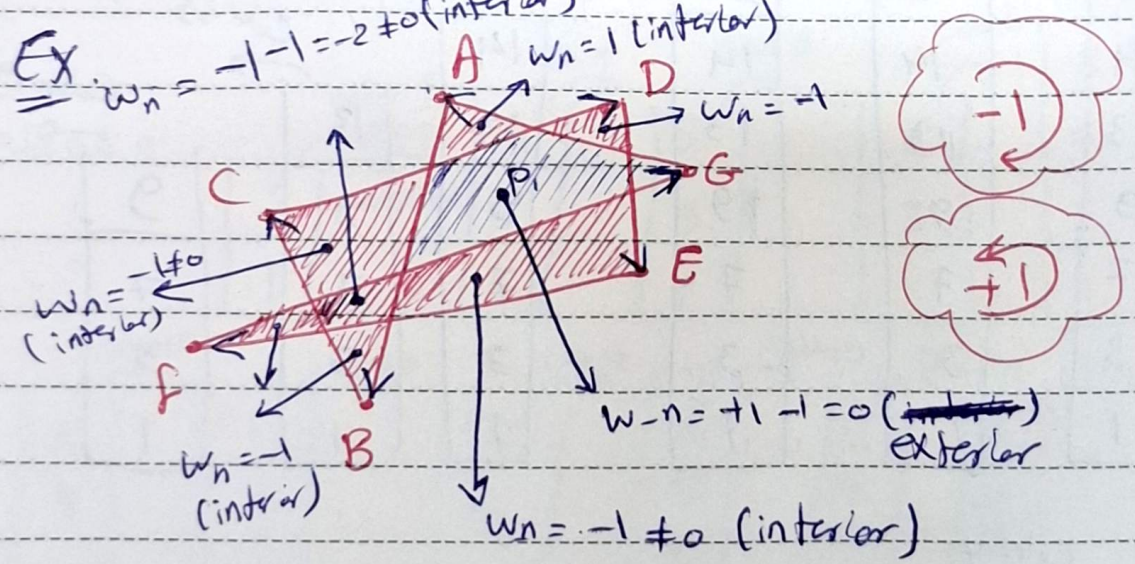
③ w-n ++ كل ما يقابل edge من اليسار لليمين

w-n -- كل ما يقابل edge من اليمين لليسار

if w-n ≠ 0 ∴ p is interior point

else (w-n = 0) p is exterior point

Ex



$w_n = -1-1 = -2 \neq 0$ (interior)

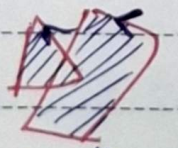
$w_n = 1$ (interior)

$w_n = -1$

$w_n = -1 \neq 0$ (interior)

$w_n = -1 \neq 0$ (interior)

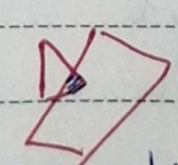
$w\text{-}n = +1-1 = 0$ (exterior)



Fill all

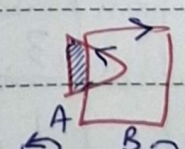
w-n ≠ 0

لون الشكل



Fill intersection

$w_n > 1$



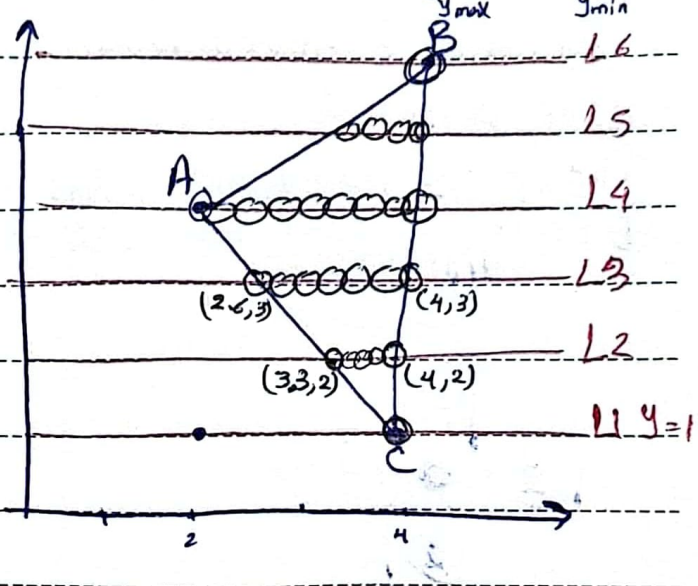
A-B

w-n > 0 — fill

DAWENHA designs

3] Scan line polygon fill :-

Ex 1) Using scanline polygon fill algorithm to determine the content of the active edge table to fill the Polygon with vertices A(2,4) B(4,6) C(4,1)
for y = 1 to 6



① ارتب الـ edges من اقل y الى y كده الترتيب في الشكل ازاي:

• C(4,1) → A(2,4) → B(4,6)

② عند كل y هتطلع عند كل Vertex الى بتقاطع معاه

③ هتعمل (GET) global edge table الاول (array of linked lists)

④ هتطلع y من min الى max

y	vertex	
6	B	→ null
5	∅	→ null
4	A	→ AB [6 | 2 | 1]
3	∅	→ null
2	∅	→ null
1	C	→ CA [4 | 4 | -2/3] → CB [6 | 4 | 0]

⑤ هتطلع من اخر واحد هتحتش عند كل Vertex ؟ لاقيت متوصل بأنهي edge

linked list الاول و اللي بعدها واللي بعدها وهكذا

* CA → (4,1) to (2,4) (x_1, y_1 to x_2, y_2)

[y_{max} = 4 | $X_{y_{min}}$ = 4 | $\frac{1}{m}$ = -2/3]

$y_2 > y_1$; X_1 for y_1 is min

$\frac{\Delta x}{\Delta y} = \frac{2-4}{4-1} = \frac{-2}{3}$

* CB → (4,1) to (4,6)

[6 | 4 | 0]

$\frac{1}{m} = \frac{4-4}{6-1} = \frac{0}{5} = 0$

* AB → (2,4) to (4,6)

[6 | 2 | 1]

$\frac{1}{m} = \frac{4-2}{6-4} = \frac{2}{2} = 1$

DAWENHA designs

③ صف active edge table بكل i مين y بطلع الـ pair

⑦ صف linked lists بعمل GET بجيب

عمل Fill بيتعم

$i=1$

| 4 | 4 | $-2/3$ | → | 6 | 4 | 0 |

pair {(4,1),(4,1)}

⑨ الصيغة $X_{k+1} = X_k + \frac{1}{m}$

⑧ بطلع الـ pair من كل list (X1,1),(X2,1)

$i=2$ no added lists

| 4 | 3.3 | $-2/3$ | → | 6 | 4 | 0 |

pair {(3.3,2),(4,2)}

⑩ بشوف لو list اضيفت بجيب list

$i=3$ no added

| 4 | 2.66 | $-2/3$ | → | 6 | 4 | 0 |

pair {(2.66,3),(4,3)}

$i=4$ add list ⑪ then sort according to X from min to max

| 4 | 1.99 | $-2/3$ | → | 6 | 2 | 1 | → | 6 | 4 | 0 |

=i

⑫ لو كان $y=i$ الـ Cancel بعمل

∴ pair = {(2,4),(4,4)}

$i=5$ no add

| 6 | 3 | 1 | → | 6 | 4 | 0 |

pair = {(3,5),(4,5)}

$i=6$ no add

| 6 | 4 | 1 | → | 6 | 4 | 0 |

pair = {(4,6),(4,6)}

$y=i$ آخر مرة

Cancel after $i=y=6$

DAWENHA designs

CamScanner

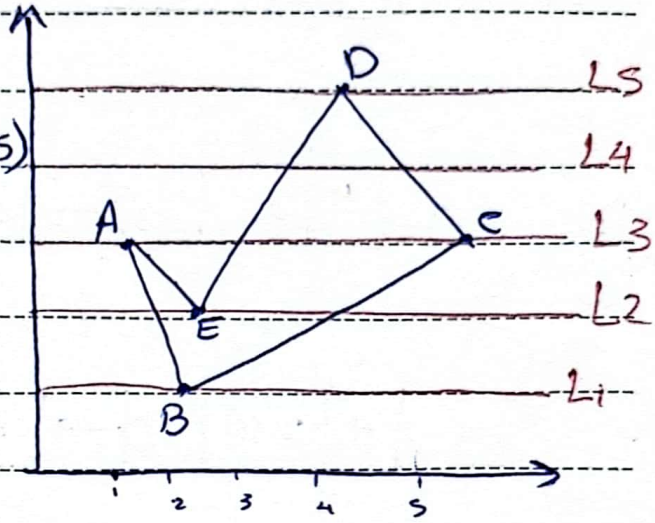
Ex. 2 Using Scanline fill to determine the content of the active edge table to fill the polygon with vertices A(1,3), B(2,1), C(5,3), D(4,5) E(2,2)

y_{min} y_{max}

y from 1 to 5

① Sort vertex:

B(2,1), E(2,2), A(1,3), C(5,3), D(4,5)



② Creat GET

y	Vertex	
5	D	→ null
4	O	→ null
3	A, C	CD → 5 | 5 | -1/2
2	E	EA → 3 | 2 | -1 ; ED → 5 | 2 | 2/3
1	B	BA → 3 | 2 | -1/2 ; BC → 3 | 2 | 3/2

~~for BA~~ → (2,1) (1,3) 3 | 2 | -1/2 $\frac{1-2}{3-1} = \frac{-1}{2}$

for BC → (2,1) (5,3) 3 | 2 | 3/2 $\frac{5-2}{3-1} = \frac{3}{2}$

for EA → (2,2) (1,3) 3 | 2 | -1 $\frac{1-2}{3-2} = -1$

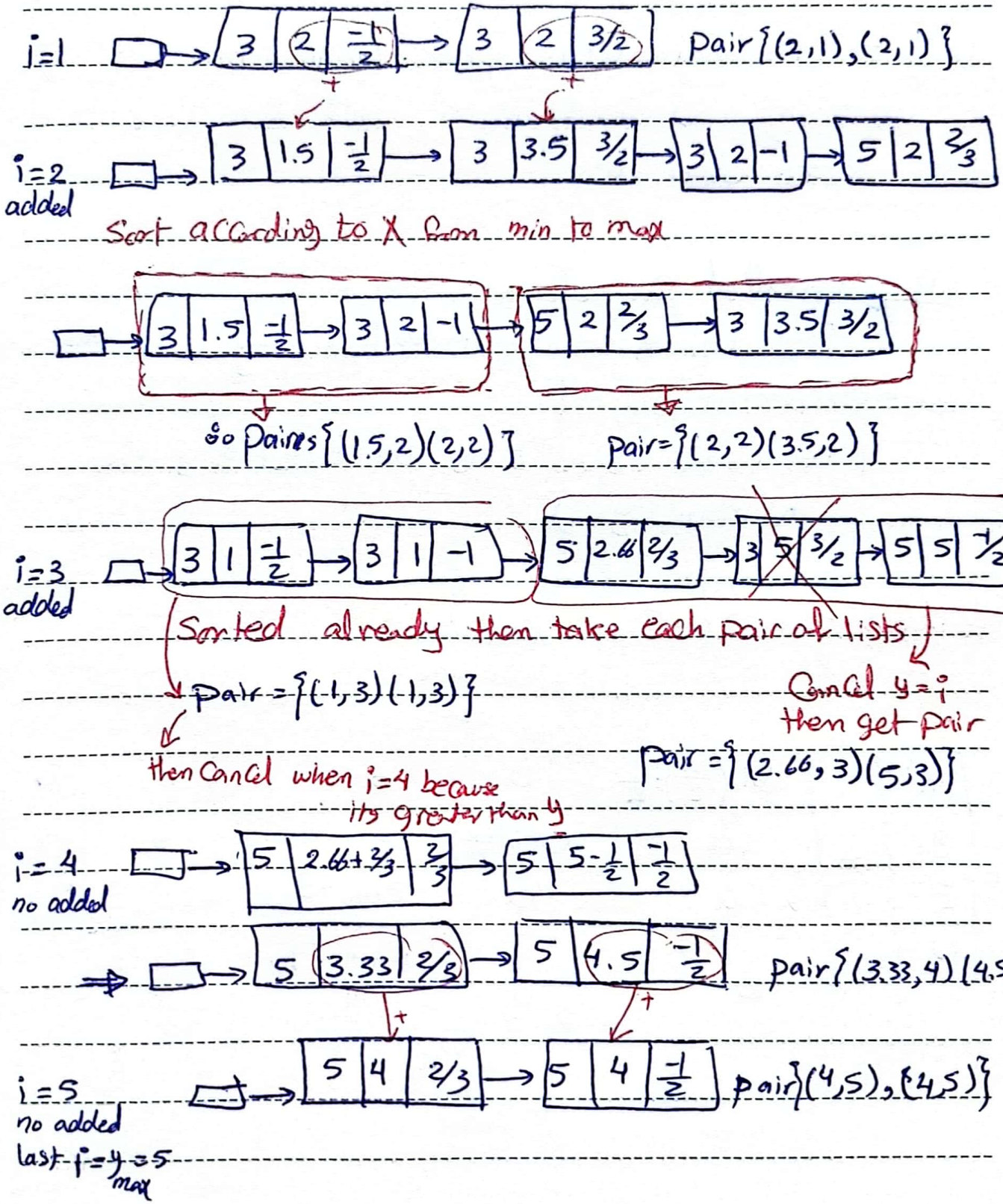
for ED → (2,2) (4,5) 5 | 2 | 2/3 $\frac{4-2}{5-2} = \frac{2}{3}$

for CD → (5,3) (4,5) 5 | 5 | -1/2 $\frac{4-5}{5-3} = \frac{-1}{2}$

DAWENHA designs

CamScanner

③ Create active edge table :-



DAWENHA designs

CamScanner

Sec 9 : projection

refer to the mathimatical process (matrix multiplication) of converting 3D Cordinates of points in a scene into screen Cordinates

1 → parallel projection
(by drawing straight lines from every point on the object)

2 → prospective projection
(represent objects as perceived by the human eyes)

* Prespective projection matrix

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} \cot\left(\frac{fov}{2}\right) * \frac{1}{a} & 0 & 0 & 0 \\ 0 & \cot\left(\frac{fov}{2}\right) & 0 & 0 \\ 0 & 0 & \frac{z_{near} + z_{far}}{z_{near} - z_{far}} & \frac{-2 z_{near} z_{far}}{z_{near} - z_{far}} \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

- $\frac{fov}{2}$: field of view angle
- a : aspect ratio
- d : field of view scaling factor
- S_z : scaling factor
- t_z : transfer factor

DAWENHA designs

CamScanner

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \xrightarrow{\text{Conversion to Screen space}} \begin{bmatrix} d * \frac{x}{a} \\ d * y \\ s_z * z - t_z \\ w \end{bmatrix}$$

CamScanner

Ex.1 project the point (2,3,5) onto a screen using a camera with a fov of 90°, aspect ratio of 16:9 & near/far planes at 1 and 100 units

(z_{near}, z_{far})

$$a = \frac{16}{9} = 1.77, \quad d = \cot\left(\frac{90}{2} * \frac{\pi}{180}\right) = 1, \quad S = \frac{1+100}{1-100} = -1.02$$

(3.14)

$$t = \frac{2*1*100}{1-100} = -2.02$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} d/a & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & S & -t \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \\ 1 \end{bmatrix}$$

$$x' = \frac{2d}{a}$$

$$y' = 3d$$

$$z' = 5S - t$$

* prefer perspective divide :

$$x' = \frac{x'}{z} = \frac{}{5} \qquad y' = \frac{y'}{5} \qquad z' = \frac{z'}{5}$$

DAWENHA designs

CamScanner